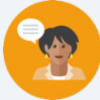


12.2 Creating Displays for Numerical Data Sets

Lesson Introduction

 Benchmarks	MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.		MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, and whether it is univariate or bivariate, and interpret the different components and quantities in the display.		 Learning Targets	- I can distinguish between univariate and bivariate data, including which representations can be used to display each type of data set. - I can create a graphical display for univariate numerical data. - I can create a graphical display for bivariate numerical data.
 Benchmark Coherence	Building On				Working Toward	
	MA.8.DP.1.1 Given a set of real-world, bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.				N/A	
 Essential Questions	- What are the similarities and differences with creating a graphical display for univariate vs bivariate numerical data? - What is the difference between univariate and bivariate data?					
 Lesson Narrative	In this lesson, students will learn to distinguish between univariate and bivariate data, including the different representations that can be used to display each type. Students will begin by getting a refresher on the various displays for bivariate numerical data. They will then work on creating displays for numerical data using directed representations. Students will then bring it together to learn how to choose the best graphical representation for numerical data.					
 Component Summary	12.2.1 Warm-Up (~5 min.)	Categorical vs. numerical, and univariate vs. bivariate classification activity (<i>SE p. 202</i>)		12.2.3 Try It (10-15 min.)	Create displays for numerical data (<i>SE p. 204</i>)	
	12.2.2 Refresher (10-15 min.)	Refresher on displays for bivariate numerical data (<i>SE p. 202</i>)		12.2.4 Bring It Together (10-15 min.)	Introduction to choosing the best graphical representation for numerical data (<i>SE p. 206</i>)	
	12.2.5 Wrap-Up (~5 min.)	Student lesson reflection (<i>SE p. 207</i>)				
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> - Line Plots (Dot Plots) - Histograms - Stem-and-Leaf Plots - Box Plots - Scatter Plots - Line Graphs <u>Additional Terms:</u> N/A		(Optional) Graphing Utility (Optional) Chart Paper & Markers (Optional) Timer		(Optional) Consider preparing a side-by-side display of scatter plots and line graphs for 12.2.2 Refresher.	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may reverse the variables when creating their scatterplot.	- The connection between 12.2.2 Refresher and 12.2.3 Try It to 12.2.4 Bring It Together is very important. In the first two components, students are exploring the individual graphs based on the situation, with prompting for the direction they should go. In 12.2.4 Bring It Together, students learn under which conditions a specific representation would be most appropriate. - The activities can vary in length of time based on student prerequisite knowledge. Consider the pacing of this lesson based on student needs.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time For 12.2.4 Bring It Together, consider leveraging this instructional routine to assist students with developing a situational understanding of each visual representation. Have students think individually about a real-world example that meets the conditions provided, and then share with different partners to refine and clarify their response.	MA.K12.MTR.2.1: Multiple Representations Throughout this lesson, students are learning about different visual representations of data. Students will express connections between concepts and representations, as well as choose a representation based on the given context or purpose.
 Differentiate Instruction	Consider a modified Frayer Model to help students conceptualize the different visual displays. Consider having students provide an example and non-example in a mathematical context, and an example and non-example in a real-world context, so that they can better understand each one.	
Lesson Notes:		

12.2.1 Warm-Up: Classifying Data

Component Overview: Students will classify the data in four different scenarios as categorical or numerical, and univariate or bivariate.



12.2.1 Warm-Up: Classifying Data

When determining the best display for a data set, the first step is to determine the type of data you are being given.

Data can be classified as categorical or numerical, and univariate (one variable) or bivariate (two variables).

1. Identify the type of data for each example in the table.
The Pew Research Center collected data from a sample of 743 teens to analyze trends in social media and technology use.

Example	Categorical or Numerical?	Univariate or Bivariate?
a. The number of people you consider close friends	☐ Categorical ☒ Numerical	☒ Univariate ☐ Bivariate
b. The number of people you consider close friends and the time spent on social media per day	☐ Categorical ☒ Numerical	☐ Univariate ☒ Bivariate
c. Your most used social media site (Twitter, Instagram, Facebook, Snapchat, YouTube, or Reddit)	☒ Categorical ☐ Numerical	☒ Univariate ☐ Bivariate
d. Your most used social media site (Twitter, Instagram, Facebook, Snapchat, YouTube, or Reddit) and what effect you would say social media has had on people your age (mostly positive, mostly negative, neutral)	☒ Categorical ☐ Numerical	☐ Univariate ☒ Bivariate



Reinforce the language and explanation when eliciting student responses to 12.2.1 Warm-Up. Have students explain what, from the example, indicated that it was numerical or categorical. Also allow students to explain not only how they know it's univariate or bivariate, but also what the variable(s) is/are in that example.

12.2.2 Refresher: Displays for Bivariate Numerical Data

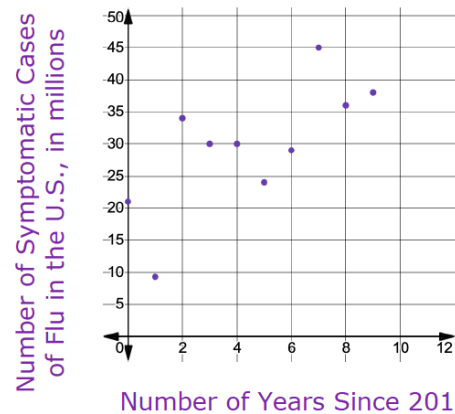
Component Overview: Students will receive a refresher on the displays for bivariate data, including both scatterplots and line graphs. For 12.2.2 Refresher, students will need a partner.



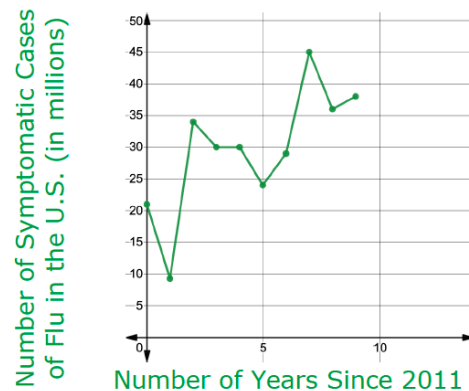
12.2.2 Refresher: Displays for Bivariate Numerical Data

Bivariate numerical data can be represented in different ways, including **scatter plots** and **line graphs**.

A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



The same data can be represented using a **line graph**.



As a launch to 12.2.2 Refresher, consider drawing a T-chart on the board with one column labeled “scatter plot,” and one column labeled “line graph.” Ask students to tell you everything they know about each graphical representation, and chart it on the board. Don’t give away any information, but use this as a springboard into the activity, and a point of reference during instruction.



To provide a visual representation for this activity, consider displaying the scatter plot and the line graph side-by-side on a virtual display, so that students can examine them side by side. Another alternative would be to create a worksheet with these two visual representations side by side, a little bit larger, so that students can explore the graphs side-by-side at their desk with their partner.



Allow students the opportunity to have some individual time to work on this before working with their partner. Consider setting a timer for individual think time, and then releasing them to work with a partner.

12.2.2 Refresher: Displays for Bivariate Numerical Data (continued)

1. Discuss with your partner the similarities and differences between the scatter plot and the line graph.

Complete the table.

Similarities Between Scatter Plots and Line Graphs	Differences Between Scatter Plots and Line Graphs
<i>Answers will vary. The labels on the axes are the same. The points plotted are the same. The scatter plot and line graph show the same data.</i>	<i>Answers will vary. On the scatter plot only the points are graphed, but on the line graph, the points are connected with a jagged line. The line graph shows the changes in the data from year to year more clearly. The scatter plot gives a better overall picture of the data.</i>

2. If data was collected on the number of symptomatic flu cases for each state for every year since 2011, which display would you choose to represent the data? Explain your reasoning.

Answers will vary. I would choose the scatterplot. With so much data, it would make sense to look for the overall trend of the data, and a scatter plot would be good for that.

3. What do you notice about the relationship between the number of symptomatic flu cases and years since 2011?

Answers will vary. Although the number of symptomatic flu cases has gone up and down since 2011, the overall trend is that they have increased.



Both scatter plots and line graphs can be used to observe the relationship between two variables in a set of bivariate data. While overall trends can be seen in both graphs, line graphs allow for closer analysis from one point to another in the data set. Scatter plots are useful for larger sets of data.



Do you think that scatter plots and line graphs can only represent bivariate data and cannot represent univariate data?

12.2.3 Try It: Creating Displays for Numerical Data

Component Overview: Students will be given three scenarios with data that they will use to create displays for.



12.2.3 Try It: Creating Displays for Numerical Data

1. The prices and size, in carats, of 20 different diamonds are shown in the table.

Weight (carats)	Actual Price (\$)
1	3,772
1	4,221
1	4,032
1	5,385
1.05	3,942
1.05	4,480
1.06	4,511
1.2	5,544
1.3	6,131
1.32	5,872
1.41	7,122
1.5	7,474
1.5	5,904
1.59	8,706
1.61	8,252
1.73	9,530
1.77	9,374
1.85	8,169
1.9	9,541
2.04	9,125

- a. The data shown in the table is ☒ numerical

and ☐ univariate.

☒ bivariate.

- b. The data shown in the table should be displayed in

a ☒ scatter plot

☐ line graph

because ...

Answers will vary. There are many points, and a scatter plot is best for that. Using a scatter plot also helps to make a generalization about the data overall and analyze the trend in price.



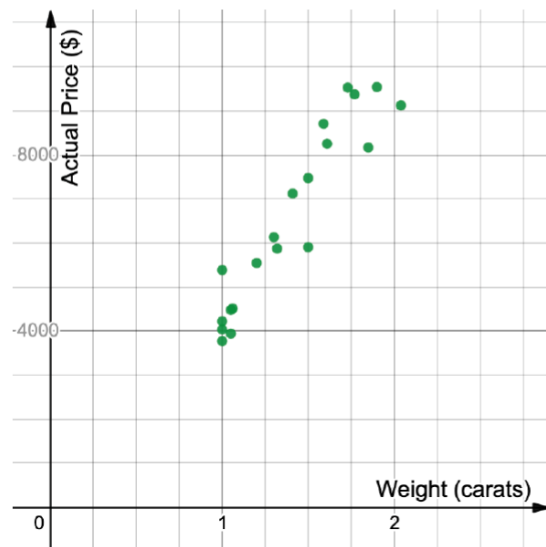
For this activity, students will be creating the visual displays for the data. During this activity, students are working in isolated scenarios and displays. They will use what they learn in this activity, coupled with the prior activity, to bring it all together in the next activity with how to choose the best display given a situation.



*How can you tell this data is numerical?
What indicates that the data is bivariate and not univariate?
Why would you choose a scatter plot and not a line graph?*

12.2.3 Try It: Creating Displays for Numerical Data (continued)

- c. Create a display to represent the relationship between size and price of a diamond.



This problem has a lot of data points. Consider allowing students to use graphing technology such as a graphing calculator, Desmos, etc. to reduce the amount of time students spend graphing this by hand.



Why is weight the label for the x -axis, and actual price the label for the y -axis?



For struggling readers, consider leveraging a word problem attack strategy to help them extract the information from the problem that is essential.

2. Laverne and Rocco planned to travel the greatest distance between any two points in the United States — from Log Point, Elliot Key, Florida to Kure Atoll, Hawaii — a distance of 5,859 miles.

They took a ferry from Log Point to Miami, then drove to Los Angeles, California, where they took a plane to Hawaii. They drove anywhere from 5-8 hours a day, and kept track of how many miles they traveled per day on their trip. This information is shown below.

{224, 294, 309, 317, 321, 339, 355, 361, 383, 456, 2500}

- a. Complete the statement.

The data set is ☒ numerical and ☒ univariate.
☐ categorical ☐ bivariate.

12.2.3 Try It: Creating Displays for Numerical Data (continued)

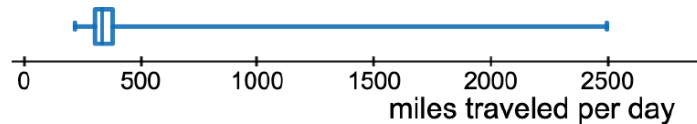
- b. Select all of the displays that **can** be used to display the data.

- ☒ Dot plot
- ☒ Histogram
- ☒ Box plot
- ☒ Stem-and-leaf
- ☐ Bar graph
- ☐ Circle graph
- ☐ Scatter plot

- c. Select all of the displays that **are best** to display the data.

- ☐ Dot plot
- ☒ Histogram
- ☒ Box plot
- ☐ Stem-and-leaf
- ☐ Bar graph
- ☐ Circle graph
- ☐ Scatter plot

- d. Create an appropriate graphical representation for the data.



- e. If the value 2500 is removed from the data set, which display would you create to represent the data? Explain your reasoning.

Answers will vary. I would create a histogram. The range of the data is still quite large, and being able to group the data into bins would help show the shape of the data better than another one of the options. However, this answer could change depending on the question that needs to be answered.



Questions 2b and 2c make for a great class discussion. Draw attention to the words in **bold** for “can” versus “best.”



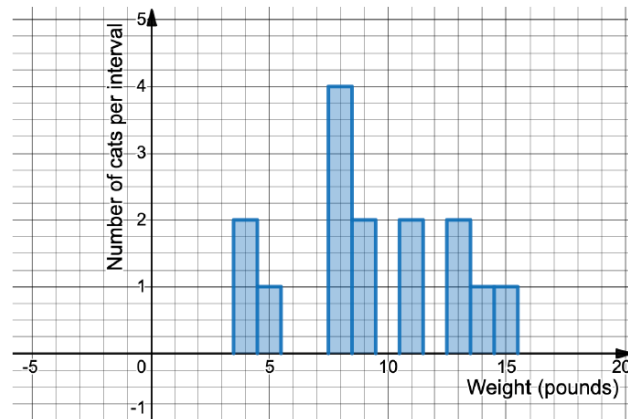
Consider showing students question 2d on a digital graphing platform so that students can see the data displayed on a larger screen. Show them what happens to the graph when 2500 is removed (part e), so students can see why another display may then become a better choice.

12.2.3 Try It: Creating Displays for Numerical Data (continued)

3. Basil recorded the weights, rounded to the nearest pound, of 15 cats at a pet show.

4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6

- a. Create an appropriate graphical display for the data. Do not use the same type of display you created in the previous problem.



- b. What other displays could be used to display his data?

Answers will vary. Another display that could be used is a line plot, box plot, or stem-and-leaf plot.



Is this data categorical or numerical? Is it univariate or bivariate? How do you know?



When students do this type of graph, they often “sketch,” not paying full attention to their marking on the x - and y -axes. Consider allowing them to create this display using a digital tool to be more accurate.

12.2.4 Bring It Together: Choosing the Best Graphical Representation for Numerical Data

Component Overview: Students will consolidate what they've learned in Lessons 1 and 2 to help them understand which scenarios are best displayed using each graphical expression.



12.2.4 Bring it Together: Choosing the Best Graphical Representation for Numerical Data

Univariate, numerical data:



Line Plots (Dot Plots)

- Best for small to moderate sets of data
- Shows the shape and distribution of data
- Shows individual data values



Histograms

- Best for larger sets of data and data over a large range,
- Can be used for continuous data
- Individual data values are not shown, only frequency over intervals
- Shows shape and distribution of the data



Stem-and-Leaf Plots

- Best for small data sets
- Shows the shape and distribution of data
- Displays individual data values in a more compressed way than a dot plot



Box Plots

- Best for large sets of data
- Displays variation of the data
- Individual data values are not shown
- Shows the 5-number summary of the data set



Consider a modified Frayer Model to help students conceptualize the different visual displays. Consider having students provide an example and non-example in a mathematical context, and an example and non-example in a real-world context, so that they can better understand each one.



Which displays are easiest for you to create by hand? Which displays are easiest for you to create using technology?

12.2.4 Bring It Together: Choosing the Best Graphical Representation for Numerical Data (continued)

Numerical, bivariate data:



Scatter plots

- Best for large sets of data
- Helpful for looking at trends over a large data set
- Useful for comparing two quantities without thinking of either one as the independent or dependent variable



Consider leveraging the “Stronger and Clearer Each Time” instructional routine to assist students with developing a situational understanding of each visual representation. Have students think individually about a real-world example that meets the conditions provided, and then share with different partners to refine and clarify their response.



Line Graphs

- Best for smaller sets of data
- Helpful for looking at small changes over time

12.2.5 Wrap-Up: Reflection

Component Overview: Students will complete a lesson reflection using two statement stems.



12.2.5 Wrap-Up: Reflection

Complete each statement based on today's lesson.

1. Today I am still unsure about ...

Answers will vary.

2. One question I still have is ...

Answers will vary.



Consider our online resources for additional remediation opportunities for students to practice or test their abilities.